Homework For Class 4: RAG with LangChain

1. Filtered retrieval across two areas

Build a retriever that searches **only** `policies/` and answer:

> *"What is the per-person limit for reimbursing a client meal, and how many days of annual leave do I get?"*

Then remove the filter and run the same question again. Print the filenames of the retrieved chunks in both cases and explain why the results differ.

Files involved: `expense_policy.pdf`, `employee_handbook.pdf`

Topics: metadata filtering, `area` field, `similarity_search` with `filter=`

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2. MMR vs plain similarity

The onboarding wiki covers VPN setup, two-factor auth, and the password manager in separate sections that are semantically close.

Ask: *"How do I keep my accounts secure?"*

Run the retriever twice — once with `search_type="similarity"` and once with `search_type="mmr"` — both at `k=4`. Print the `h2` heading from each retrieved chunk's metadata. Explain in a comment which set of results is more useful and why.

Files involved: `onboarding.md`

Topics: `MarkdownHeaderTextSplitter` metadata, MMR diversity, retriever `search_type`

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3. Streaming answer with live source list

Write a function `stream_answer(question: str)` that:

1. Retrieves chunks with the existing `retriever`
2. Prints each source filename on its own line before streaming starts

3. Streams the answer token-by-token using `rag_chain.stream()`

Call it with: *"What happens if I miss the on-call pager alert?"*

Files involved: `onboarding.md`

Topics: `.stream()`, sources from retriever, combining explicit and LCEL steps

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4. Swap the embedding provider

Replace `OpenAIEmbeddings` with `HuggingFaceEmbeddings` (`sentence-transformers/all-MiniLM-L6-v2`, 384 dims) and rebuild the in-memory vector store from the same `chunks`.

Ask the same question from Part 4 (*"How many vacation days do I get?"*) with both stores and print:

- the top result's filename and score from each
- the vector dimension of each model

Write one sentence noting any difference in the retrieved chunks.

Files involved: full corpus

Topics: embedding provider swap, dimension mismatch consequence, `similarity_search_with_score`

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5. Question the support FAQ

The `support_faq.csv` has 63 rows loaded as one Document per row. Build a **separate** in-memory vector store from only those 63 chunks (no other files) and answer:

> *"My order was cancelled — when will I get my money back?"*

Print the top 3 results with their `category` metadata. Then change `k` from 3 to 1 and note whether the single best result is enough to answer the question fully.

Files involved: `support_faq.csv`

Topics: `CSVLoader` with `metadata_columns`, isolated vector store, choosing `k`